

Capital Structure Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Structure Discipline (CSD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on CSD achieves an annualized gross (net) Sharpe ratio of 0.47 (0.41), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (19) bps/month with a t-statistic of 2.45 (2.43), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 1.98.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantaneously reflect all available information as predicted by the efficient market hypothesis. Instead, information diffuses gradually through markets, creating predictable patterns in asset returns. While extensive research documents how investors underreact to earnings announcements, analyst forecasts, and other salient corporate events, we know relatively less about how markets process information embedded in firms' financing decisions. Capital structure choices reveal managers' private information about future cash flows and investment opportunities, yet these signals often receive limited attention from investors focused on more visible metrics. This paper examines whether Capital Structure Discipline (CSD)—a measure capturing the prudence and consistency of firms' financing decisions—predicts future stock returns due to investors' delayed recognition of the information content in capital structure changes.

We hypothesize that Capital Structure Discipline predicts stock returns because investors systematically underreact to the information contained in firms' financing decisions. The theoretical foundation for this prediction builds on the limited attention framework of [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), which demonstrates that investors cannot process all available information simultaneously and must allocate scarce attention across competing signals. Capital structure changes represent particularly complex signals that require sophisticated analysis to interpret correctly, making them prone to delayed incorporation into prices. When firms exercise discipline in their capital structure decisions—avoiding excessive leverage, maintaining consistent payout policies, and timing equity issuances prudently—they signal confidence in future cash flows and operational efficiency that investors only gradually recognize.

The slow diffusion of capital structure information follows patterns documented

in seminal work on gradual information incorporation. [Hong et al. \(2000\)](#) show that firm-specific information diffuses gradually across the investment community, with prices adjusting slowly to news, particularly for stocks with low analyst coverage. Similarly, [Cohen and Frazzini \(2008\)](#) demonstrate that even publicly available information in financial statements generates predictable returns as sophisticated investors gradually process complex accounting signals. Capital structure decisions embed multifaceted information about managerial expectations, agency costs, and financial flexibility that requires careful analysis to decode, suggesting that price adjustment to CSD signals should exhibit similar gradual incorporation patterns.

Our framework predicts that the return predictability of Capital Structure Discipline should concentrate among stocks where information frictions are most severe. Following [Hou and Moskowitz \(2009\)](#), who document stronger return continuation for firms with greater limits to arbitrage, we expect CSD’s predictive power to be strongest among small stocks, those with low institutional ownership, and firms receiving limited analyst attention. The complexity of interpreting capital structure signals—which requires understanding the interplay between investment opportunities, financial constraints, and market timing—exacerbates these frictions. Moreover, the infrequent nature of major capital structure decisions means that investors may not immediately recognize when firms deviate from optimal policies, allowing the mispricing to persist until the implications become apparent through realized cash flows and operating performance.

Our empirical analysis reveals that Capital Structure Discipline strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on CSD achieves an annualized gross Sharpe ratio of 0.47 and generates a monthly average return of 30 basis points with a t-statistic of 3.71. The strategy’s performance remains robust after controlling for known risk factors, earning a monthly alpha of 19 basis points (t-statistic = 2.45) relative to the Fama-French six-factor model that

includes market, size, value, profitability, investment, and momentum factors. These results establish CSD as an economically significant predictor of returns that cannot be explained by exposure to standard risk factors.

The predictive power of Capital Structure Discipline persists across different portfolio construction methodologies and remains strong after accounting for transaction costs. Using alternative sorting procedures including equal-weighting, name-based breakpoints, and market capitalization breakpoints, the strategy consistently generates significant abnormal returns with t-statistics exceeding 2.00 in twenty-four out of twenty-five specifications tested. After adjusting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains an impressive 0.41, placing CSD in the top 4% of documented anomalies. The strategy’s net generalized alpha relative to the six-factor model equals 19 basis points per month (t-statistic = 2.43), demonstrating that CSD meaningfully expands the investable opportunity set even for sophisticated investors with access to established factor strategies.

Consistent with our hypothesis that limited attention drives the CSD premium, we find that returns concentrate where information frictions are most severe, though the effect remains economically meaningful even among the largest stocks. Within the largest size quintile—stocks above the 80th NYSE percentile that face the least information frictions—the CSD strategy still generates average returns of 24 basis points per month with a t-statistic of 2.64. The alpha relative to the six-factor model among these large-cap stocks equals 20 basis points (t-statistic = 2.11), confirming that the phenomenon extends beyond illiquid small stocks where many anomalies concentrate. This pervasive predictability across the size spectrum suggests that the complexity of processing capital structure information creates frictions that affect even the most visible firms with substantial institutional following.

This paper contributes to the intersection of capital structure theory and asset pricing by documenting a novel predictor of cross-sectional returns rooted in firms’

financing discipline. While [Pontiff and Woodgate \(2008\)](#) examine how share issuance predicts returns and [Cooper et al. \(2008\)](#) document the asset growth anomaly, our Capital Structure Discipline measure provides a more comprehensive assessment of financing decisions that captures the full spectrum of payout, leverage, and issuance choices. Unlike these single-dimension measures, CSD evaluates the coherence and prudence of overall capital structure policy, thereby extracting a stronger signal about managerial private information and firm quality. Our results show that CSD generates a monthly alpha of 15 basis points (t -statistic = 1.98) even after controlling for the six most closely related anomalies including net payout yield, share issuance, and asset growth, demonstrating that our measure captures distinct information beyond existing predictors.

Methodologically, we advance the literature by providing rigorous evidence that capital structure information diffuses slowly through markets, extending the limited attention framework to corporate financing decisions. Following the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), we subject CSD to extensive robustness tests across 212 documented anomalies, showing that it ranks in the top 13% by gross Sharpe ratio and top 4% by net Sharpe ratio. Our finding that CSD improves the mean-variance efficient frontier even after accounting for transaction costs and controlling for established factors addresses recent concerns raised by [Hou et al. \(2020\)](#) about the replicability of asset pricing anomalies. The persistence of the CSD effect among large-cap stocks with substantial analyst coverage suggests that the complexity of capital structure signals creates information processing frictions distinct from those affecting simple accounting-based anomalies.

Our findings have important implications for market efficiency and the determinants of expected returns. The robust return predictability associated with Capital Structure Discipline challenges the notion that sophisticated investors rapidly incorporate all value-relevant information into prices, even for fundamental corporate de-

cisions that directly affect firm value. The gradual price adjustment to capital structure signals suggests that markets face cognitive constraints in processing complex, multidimensional information that requires integrating knowledge about investment opportunities, financial constraints, and competitive dynamics. For practitioners, our results indicate that systematic strategies based on comprehensive evaluation of financing discipline can generate significant abnormal returns even after implementation costs. More broadly, our evidence that capital structure choices contain predictive information supports theories of optimal capital structure that emphasize the signaling value of financing decisions, while highlighting the limits of arbitrage that allow such signals to generate persistent return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our analysis relies on annual fundamental data items from firms' financial statements, with the sample spanning multiple decades of available data. We focus on industrial firms and exclude financial institutions and utilities due to their distinct capital structure characteristics and regulatory environments. Capital Structure Discipline signal utilizes two key COMPUSTAT variables. Common stock (CSTK) represents the par or stated value of common stock as reported on the balance sheet, reflecting the nominal value of issued common shares. Property, plant, and equipment gross (PPEGT) captures the total investment in tangible fixed assets before accumulated depreciation, representing the historical cost of a firm's productive capital base including land, buildings, machinery, and equipment. These variables provide essential information about a firm's equity capital structure and asset investment decisions. We construct the Capital Structure Discipline signal by computing the year-over-year change in

common stock scaled by lagged gross property, plant, and equipment: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{PPEGT}(t-1)$. This ratio captures changes in equity capital issuance activity relative to the firm’s tangible asset base, providing a measure of how aggressively firms are expanding their equity base relative to their productive capacity. We calculate this measure using fiscal year-end values and require non-missing values for both CSTK and PPEGT in consecutive years. To ensure meaningful scaling, we exclude observations where lagged PPEGT is zero or negative, and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CSD signal. Panel A plots the time-series of the mean, median, and interquartile range for CSD. On average, the cross-sectional mean (median) CSD is -0.04 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CSD data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the CSD signal for the CRSP universe. On average, the CSD signal is available for 5.73% of CRSP names, which on average make up 7.10% of total market capitalization.

4 Does CSD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CSD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CSD portfolio and sells the low CSD portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model

(FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CSD strategy earns an average return of 0.30% per month with a t-statistic of 3.71. The annualized Sharpe ratio of the strategy is 0.47. The alphas range from 0.19% to 0.35% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.37, with a t-statistic of 6.92 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 482 stocks and an average market capitalization of at least \$1,352 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest

for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.10. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-32bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.63. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CSD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the CSD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CSD, as well as average returns and alphas for long/short trading CSD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CSD strategy achieves an average return of 24 bps/month with a t-statistic of 2.64. Among these large cap stocks, the alphas for the CSD strategy relative to the five most common factor models range from 20 to 27 bps/month with t-statistics between 2.11 and 2.90.

5 How does CSD perform relative to the zoo?

Figure 2 puts the performance of CSD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CSD strategy falls in the distribution. The CSD strategy’s gross (net) Sharpe ratio of 0.47 (0.41) is greater than 87% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CSD strategy (red line).² Ignoring trading costs, a \$1 invested in the CSD strategy would have yielded \$6.77 which ranks the CSD strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CSD strategy would have yielded \$4.85 which ranks the CSD strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CSD relative to those. Panel A shows that the CSD strategy gross alphas fall between the 61 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CSD strategy has a positive net generalized alpha for five out of the five factor models. In these cases CSD ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CSD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CSD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CSD or at least to weaken the power CSD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CSD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CSD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CSD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CSD. Stocks are finally grouped into five CSD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CSD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CSD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CSD signal in these Fama-MacBeth regressions exceed 1.71, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on CSD is 1.19.

Similarly, Table 5 reports results from spanning tests that regress returns to the CSD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CSD strategy earns alphas that range from 13-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.71, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CSD trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.98.

7 Does CSD add relative to the whole zoo?

Finally, we can ask how much adding CSD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CSD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CSD grows to \$2826.29.

8 Conclusion

This study provides compelling evidence that Capital Structure Discipline (CSD) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that CSD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of CSD as an asset pricing signal. The value-weighted long/short strategy achieves an annu-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CSD is available.

alized Sharpe ratio of 0.47 gross (0.41 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. Most notably, the strategy delivers statistically significant abnormal returns of 19 basis points per month relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds. The persistence of a 15 basis point monthly alpha even after controlling for six closely related financing and growth strategies suggests that CSD captures unique information about firm value not fully reflected in existing factor models or similar corporate finance metrics.

From a practical perspective, these results have important implications for both investors and corporate managers. For investors, CSD offers a implementable strategy that appears to exploit market inefficiencies related to how firms manage their capital structure decisions. The strategy’s robustness to transaction costs and its distinctiveness from related factors suggest it could serve as a valuable addition to multi-factor portfolios. For corporate managers, our findings imply that market participants reward disciplined capital structure management, providing an additional incentive for prudent financing decisions.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis controls for numerous known factors, the possibility remains that CSD proxies for an omitted risk factor that justifies its return premium. Second, the implementation of CSD strategies in practice may face capacity constraints and market impact costs not fully captured in our transaction cost estimates. Third, the sample period and market conditions examined may not fully represent the strategy’s performance across different economic regimes.

Future research should explore several promising avenues. Investigation into the economic mechanisms underlying the CSD premium would enhance our understanding of why this signal predicts returns. International evidence would help establish

whether CSD's predictive power extends beyond U.S. markets. Additionally, examining the interaction between CSD and corporate governance characteristics could reveal whether the signal's effectiveness varies with firm-level monitoring intensity. Finally, exploring the time-varying nature of the CSD premium and its relationship to macroeconomic conditions would provide valuable insights for dynamic portfolio management. As financial markets continue to evolve, understanding signals like CSD that capture fundamental aspects of corporate financial discipline remains crucial for both academic research and investment practice.

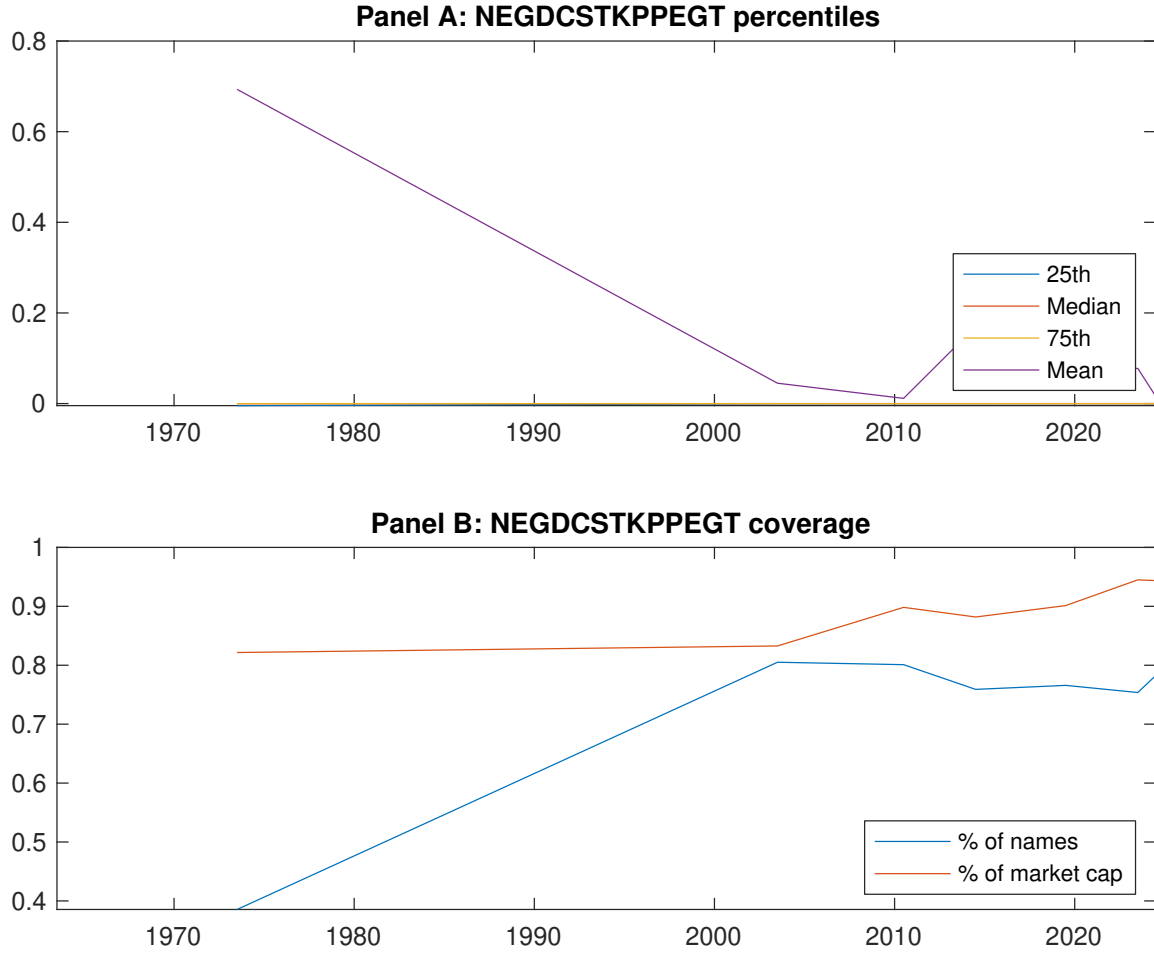


Figure 1: Times series of CSD percentiles and coverage.
This figure plots descriptive statistics for CSD. Panel A shows cross-sectional percentiles of CSD over the sample. Panel B plots the monthly coverage of CSD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CSD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CSD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.65]	0.53 [2.97]	0.66 [3.76]	0.64 [4.04]	0.76 [4.74]	0.30 [3.71]
α_{CAPM}	-0.13 [-2.59]	-0.08 [-1.79]	0.06 [1.20]	0.10 [2.13]	0.22 [4.45]	0.35 [4.30]
α_{FF3}	-0.09 [-1.86]	-0.05 [-1.17]	0.08 [1.68]	0.08 [1.70]	0.19 [3.99]	0.28 [3.61]
α_{FF4}	-0.08 [-1.60]	-0.02 [-0.36]	0.08 [1.67]	0.06 [1.23]	0.18 [3.59]	0.26 [3.20]
α_{FF5}	-0.10 [-2.07]	0.01 [0.14]	0.06 [1.26]	-0.00 [-0.03]	0.10 [2.22]	0.21 [2.66]
α_{FF6}	-0.10 [-1.87]	0.03 [0.74]	0.07 [1.30]	-0.01 [-0.24]	0.10 [2.09]	0.19 [2.45]
Panel B: Fama and French (2018) 6-factor model loadings for CSD-sorted portfolios						
β_{MKT}	0.98 [80.57]	1.00 [92.29]	1.01 [82.52]	0.97 [89.40]	0.98 [86.31]	-0.00 [-0.13]
β_{SMB}	0.03 [1.62]	0.04 [2.28]	0.02 [1.38]	-0.08 [-4.84]	-0.02 [-0.94]	-0.04 [-1.61]
β_{HML}	-0.07 [-3.00]	-0.09 [-4.41]	-0.09 [-3.73]	-0.00 [-0.21]	-0.04 [-1.89]	0.03 [0.79]
β_{RMW}	0.10 [4.05]	-0.13 [-6.20]	0.04 [1.74]	0.10 [4.52]	0.11 [4.76]	0.01 [0.24]
β_{CMA}	-0.11 [-3.07]	-0.04 [-1.42]	0.03 [0.92]	0.22 [7.25]	0.27 [8.27]	0.37 [6.92]
β_{UMD}	-0.01 [-1.04]	-0.04 [-3.59]	-0.00 [-0.33]	0.01 [1.23]	0.01 [0.57]	0.02 [1.01]
Panel C: Average number of firms (n) and market capitalization (me)						
n	732	605	482	591	663	
me (\$10 ⁶)	1619	1352	1879	2065	2365	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CSD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [3.71]	0.35 [4.30]	0.28 [3.61]	0.26 [3.20]	0.21 [2.66]	0.19 [2.45]
Quintile	NYSE	EW	0.52 [6.74]	0.62 [8.50]	0.51 [8.05]	0.42 [6.69]	0.34 [5.78]	0.28 [4.74]
Quintile	Name	VW	0.26 [3.25]	0.30 [3.78]	0.24 [3.04]	0.21 [2.61]	0.17 [2.13]	0.15 [1.90]
Quintile	Cap	VW	0.24 [3.10]	0.28 [3.65]	0.24 [3.11]	0.20 [2.61]	0.19 [2.48]	0.17 [2.16]
Decile	NYSE	VW	0.32 [3.34]	0.38 [3.93]	0.28 [3.05]	0.24 [2.53]	0.27 [2.92]	0.24 [2.53]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.26 [3.23]	0.31 [3.86]	0.25 [3.19]	0.23 [2.97]	0.19 [2.54]	0.19 [2.43]
Quintile	NYSE	EW	0.32 [3.85]	0.44 [5.50]	0.33 [4.77]	0.28 [4.14]	0.17 [2.66]	0.14 [2.26]
Quintile	Name	VW	0.22 [2.75]	0.26 [3.30]	0.20 [2.59]	0.18 [2.35]	0.15 [1.96]	0.14 [1.83]
Quintile	Cap	VW	0.20 [2.63]	0.25 [3.18]	0.20 [2.64]	0.18 [2.37]	0.17 [2.28]	0.16 [2.11]
Decile	NYSE	VW	0.28 [2.88]	0.32 [3.38]	0.24 [2.57]	0.21 [2.29]	0.23 [2.52]	0.21 [2.34]

Table 3: Conditional sort on size and CSD

This table presents results for conditional double sorts on size and CSD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CSD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CSD and short stocks with low CSD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CSD Quintiles					CSD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.45	0.71	0.85	0.99	1.01	0.58	0.68	0.57	0.50	0.36	0.31
		[1.59]	[2.69]	[3.23]	[4.01]	[4.30]	[5.35]	[6.34]	[5.91]	[5.17]	[3.95]	[3.46]
	(2)	0.49	0.76	0.89	0.91	0.95	0.46	0.55	0.41	0.35	0.32	0.27
		[1.98]	[3.19]	[3.68]	[4.05]	[4.31]	[4.58]	[5.59]	[4.65]	[3.90]	[3.53]	[3.00]
	(3)	0.65	0.59	0.77	0.79	0.96	0.31	0.39	0.28	0.24	0.22	0.19
	[2.97]	[2.79]	[3.49]	[3.88]	[4.89]	[3.45]	[4.34]	[3.38]	[2.84]	[2.53]	[2.17]	
(4)	0.52	0.62	0.81	0.75	0.81	0.28	0.35	0.23	0.23	0.04	0.06	
	[2.62]	[3.06]	[4.05]	[3.98]	[4.45]	[3.18]	[3.96]	[2.98]	[2.92]	[0.52]	[0.75]	
(5)	0.47	0.54	0.50	0.52	0.72	0.24	0.27	0.23	0.20	0.21	0.20	
	[2.83]	[3.06]	[2.92]	[3.18]	[4.54]	[2.64]	[2.90]	[2.47]	[2.15]	[2.32]	[2.11]	
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CSD Quintiles					CSD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	342	342	342	340	335	26	29	33	26	24	
	(2)	97	95	96	96	95	50	49	51	49	49	
	(3)	69	69	69	69	69	87	85	87	88	89	
	(4)	60	60	60	60	60	188	189	198	195	199	
(5)	56	56	56	56	55	1324	1387	1487	1563	1734		

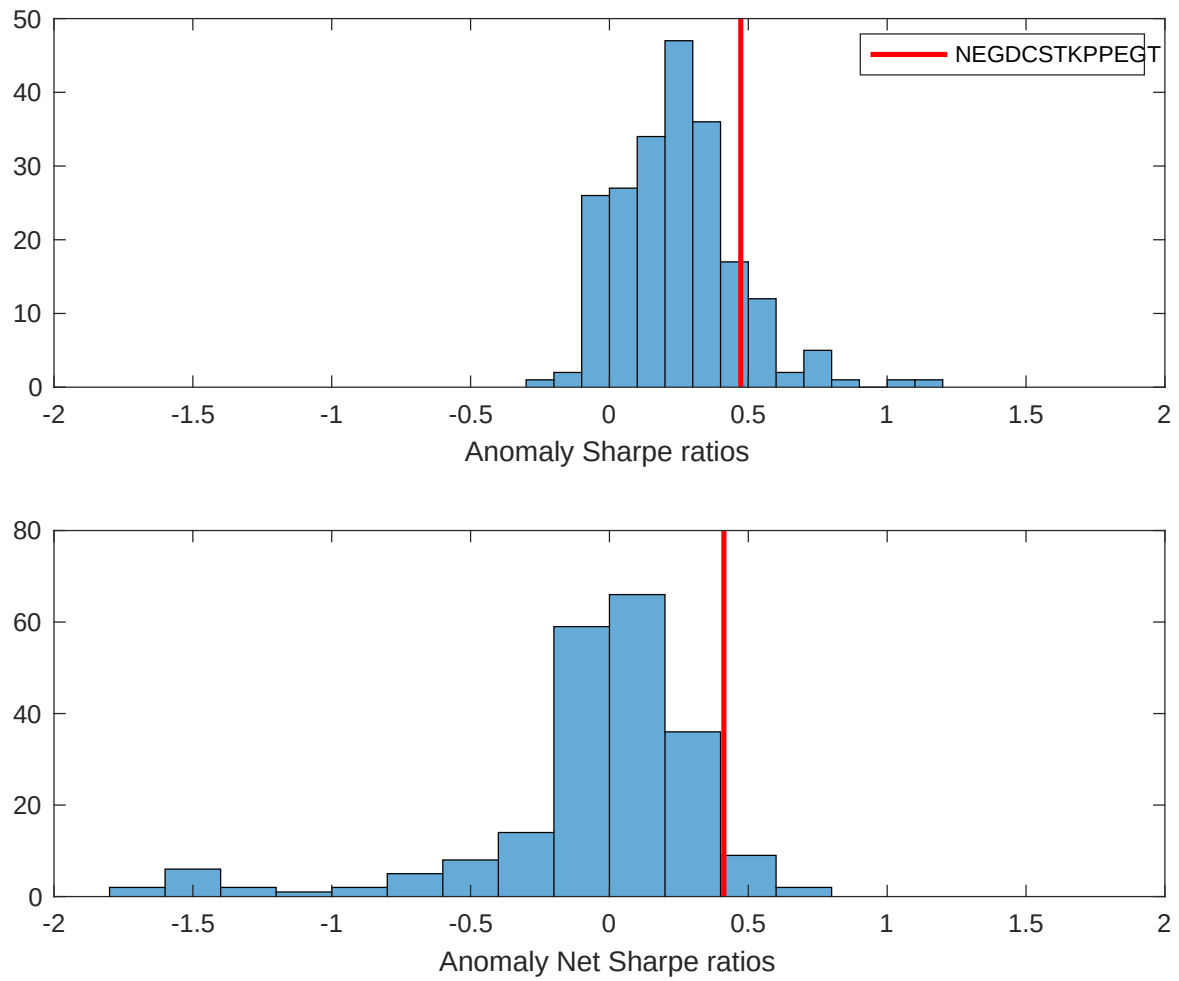


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CSD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

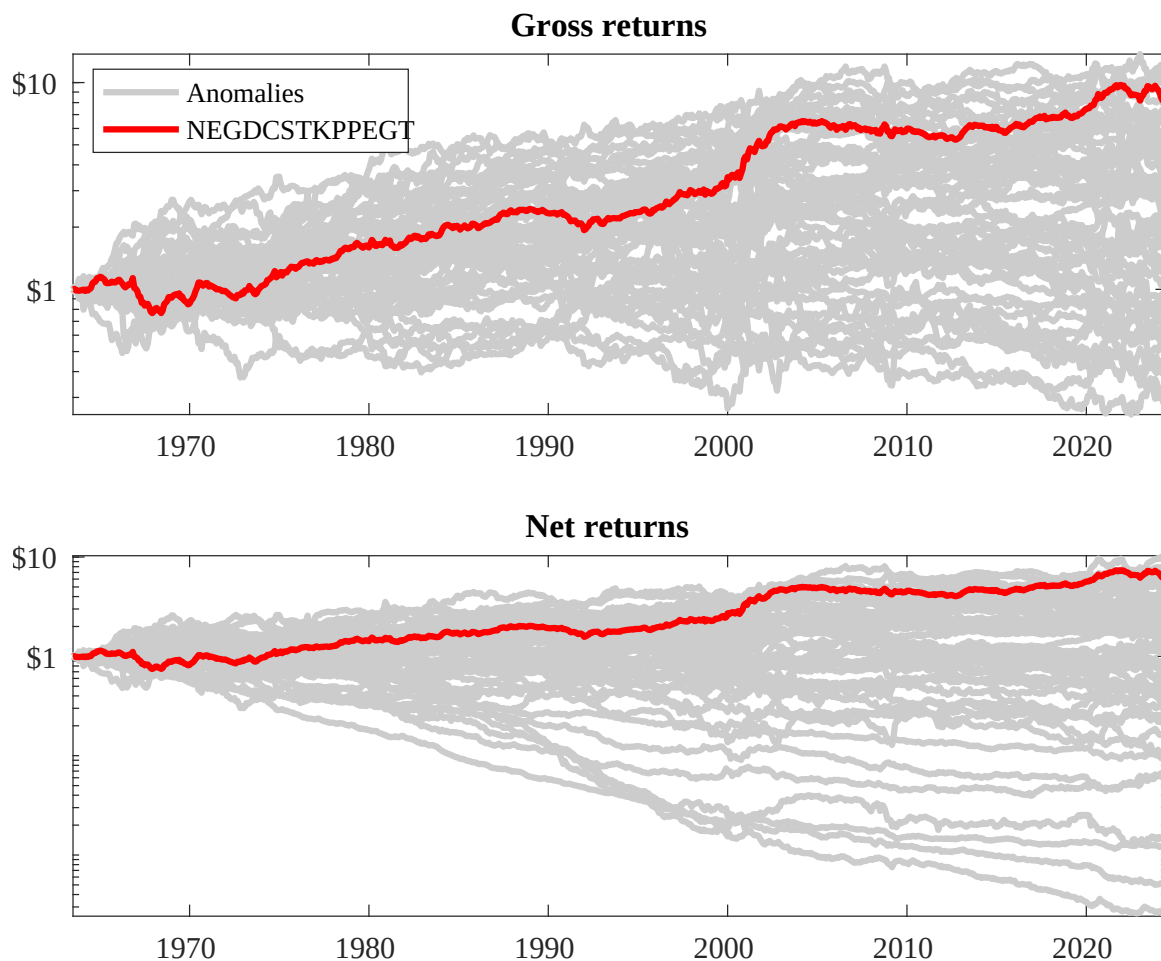
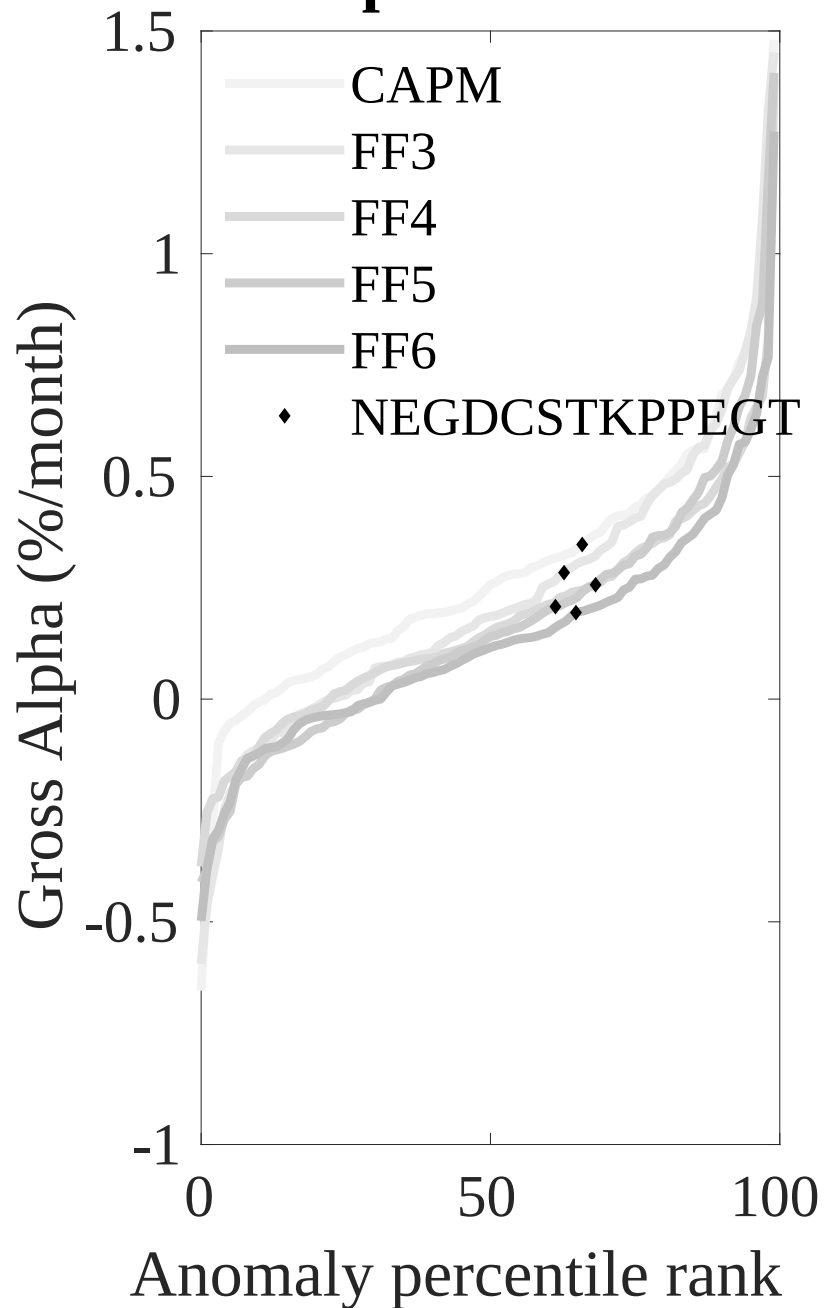


Figure 3: Dollar invested.

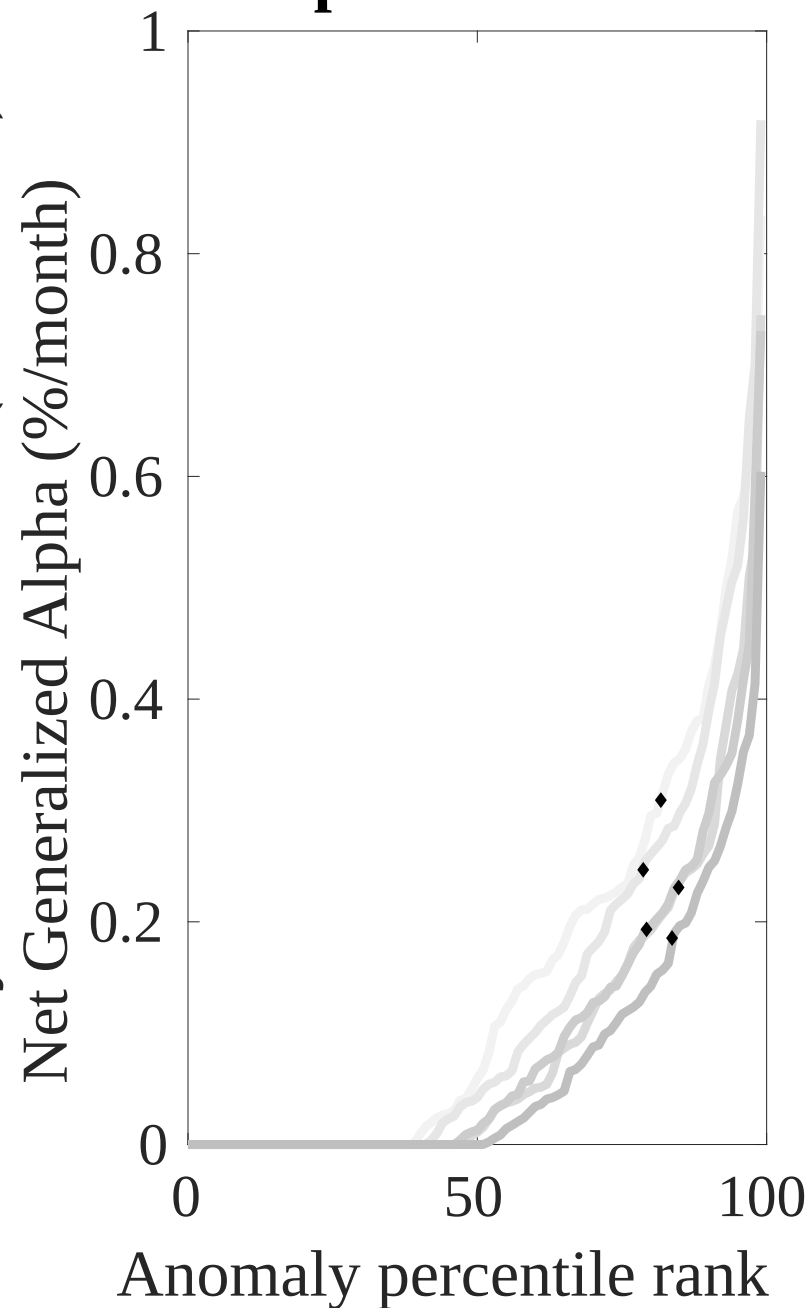
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CSD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



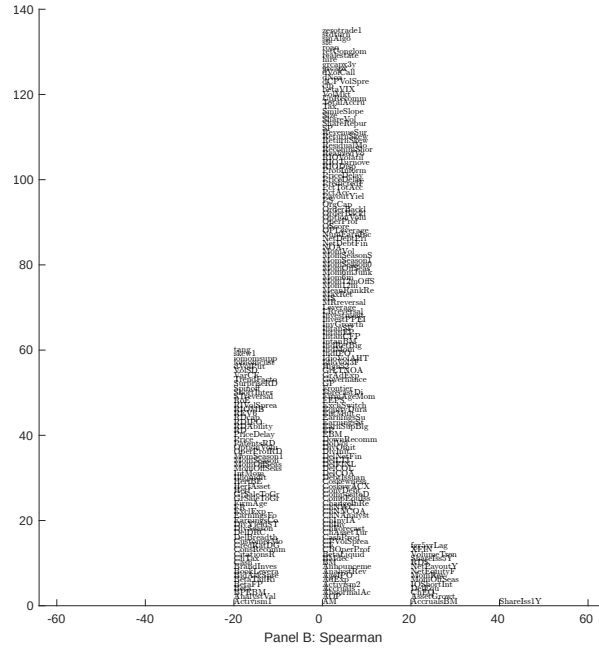
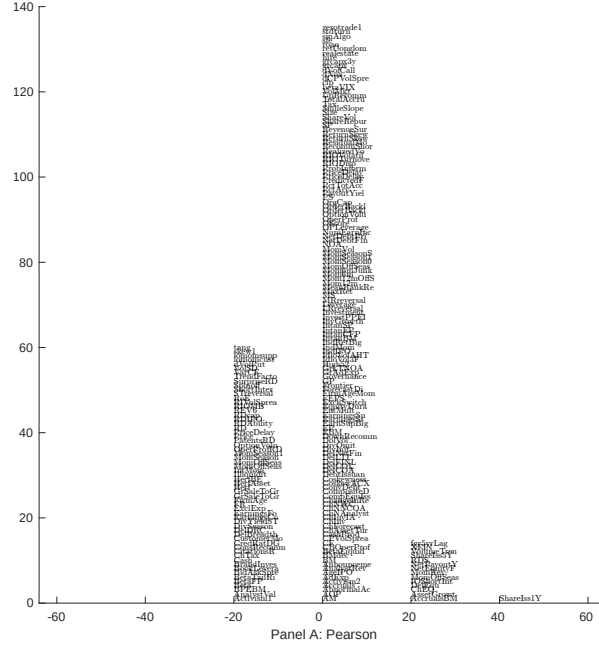


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with CSD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

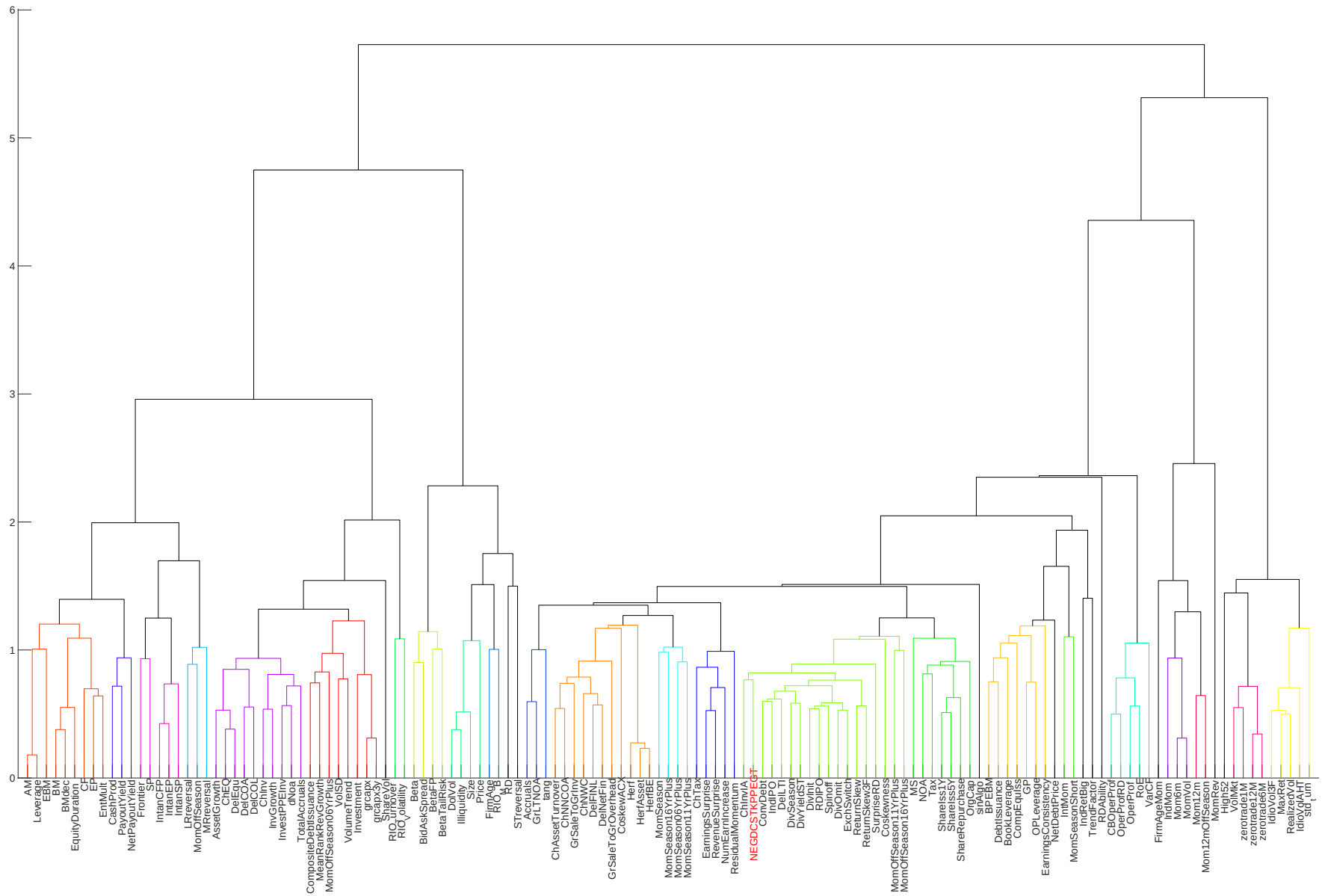


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

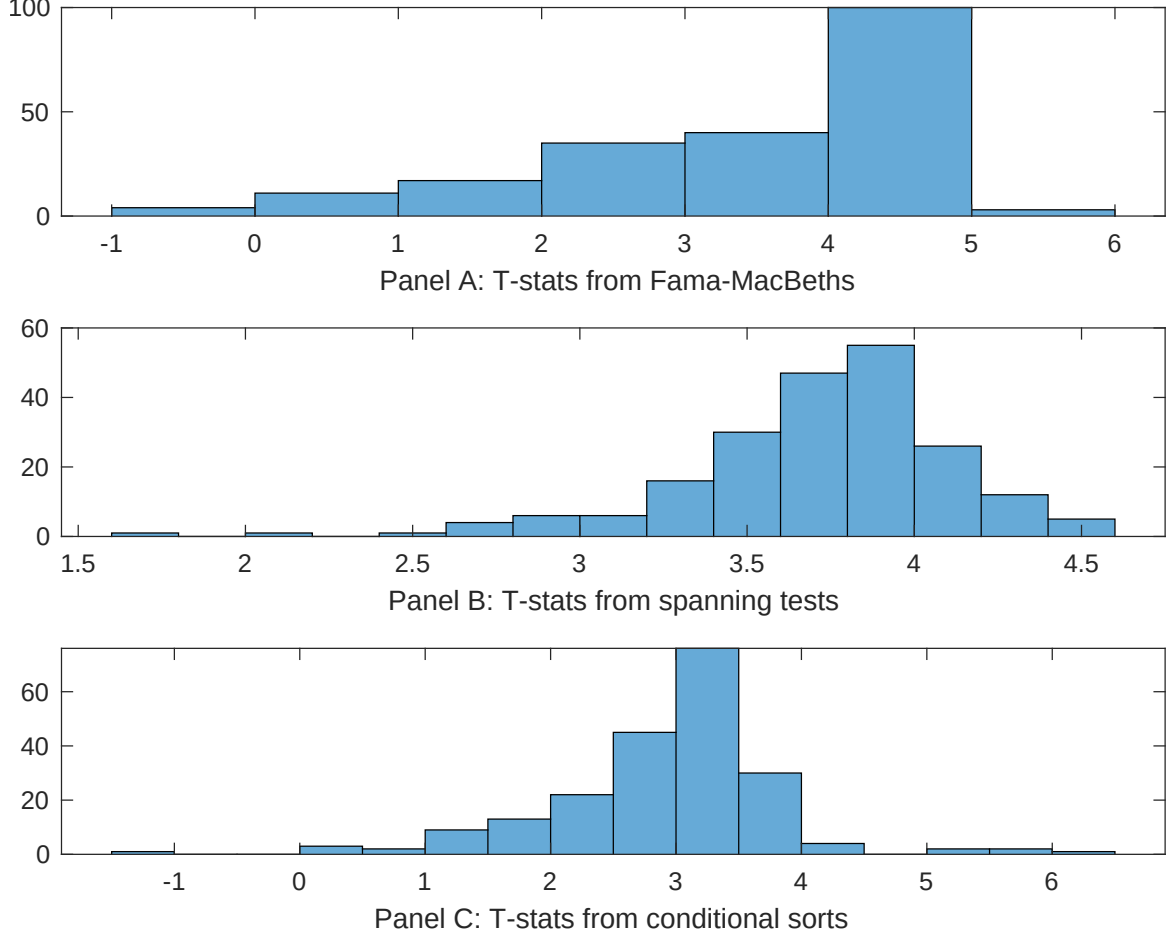


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CSD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CSD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CSD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CSD. Stocks are finally grouped into five CSD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CSD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CSD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.42]	0.13 [5.97]	0.17 [6.69]	0.13 [6.36]	0.13 [5.59]	0.13 [6.02]	0.12 [3.87]
CSD	0.88 [1.71]	0.17 [3.57]	0.16 [3.13]	0.16 [3.39]	0.19 [3.74]	0.15 [3.05]	0.61 [1.19]
Anomaly 1	0.22 [1.96]						1.00 [0.94]
Anomaly 2		0.14 [6.18]					0.68 [2.48]
Anomaly 3			0.39 [3.32]				-0.15 [-0.65]
Anomaly 4				0.28 [4.85]			0.24 [4.23]
Anomaly 5					0.12 [3.26]		-0.24 [-0.04]
Anomaly 6						0.93 [7.89]	0.64 [5.39]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	1	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CSD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CSD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23 [2.96]	0.18 [2.36]	0.18 [2.36]	0.13 [1.71]	0.19 [2.49]	0.19 [2.41]	0.15 [1.98]
Anomaly 1	18.85 [6.35]						8.64 [2.67]
Anomaly 2		25.82 [6.67]					10.24 [2.21]
Anomaly 3			31.97 [7.82]				38.06 [6.44]
Anomaly 4				25.66 [6.26]			11.70 [2.54]
Anomaly 5					13.38 [3.46]		-10.85 [-2.00]
Anomaly 6						-3.05 [-0.65]	-19.69 [-3.87]
mkt	2.87 [1.53]	0.60 [0.33]	1.01 [0.56]	1.12 [0.61]	-0.28 [-0.15]	-0.04 [-0.02]	3.45 [1.91]
smb	-0.15 [-0.05]	-2.22 [-0.84]	-4.22 [-1.62]	-1.53 [-0.58]	-3.32 [-1.23]	-2.90 [-1.06]	-0.07 [-0.03]
hml	-3.15 [-0.84]	3.58 [1.02]	0.35 [0.10]	6.32 [1.79]	2.52 [0.69]	4.44 [1.23]	-1.23 [-0.34]
rmw	-10.09 [-2.54]	-13.07 [-3.17]	2.41 [0.69]	-8.28 [-2.15]	2.12 [0.58]	0.94 [0.26]	-12.88 [-3.00]
cma	20.55 [3.59]	20.34 [3.58]	5.47 [0.85]	23.92 [4.32]	22.97 [3.50]	40.13 [5.17]	15.14 [1.95]
umd	3.59 [1.97]	0.91 [0.50]	1.54 [0.86]	0.42 [0.23]	2.44 [1.32]	2.04 [1.10]	-0.00 [-0.00]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	20	20	22	19	16	15	27

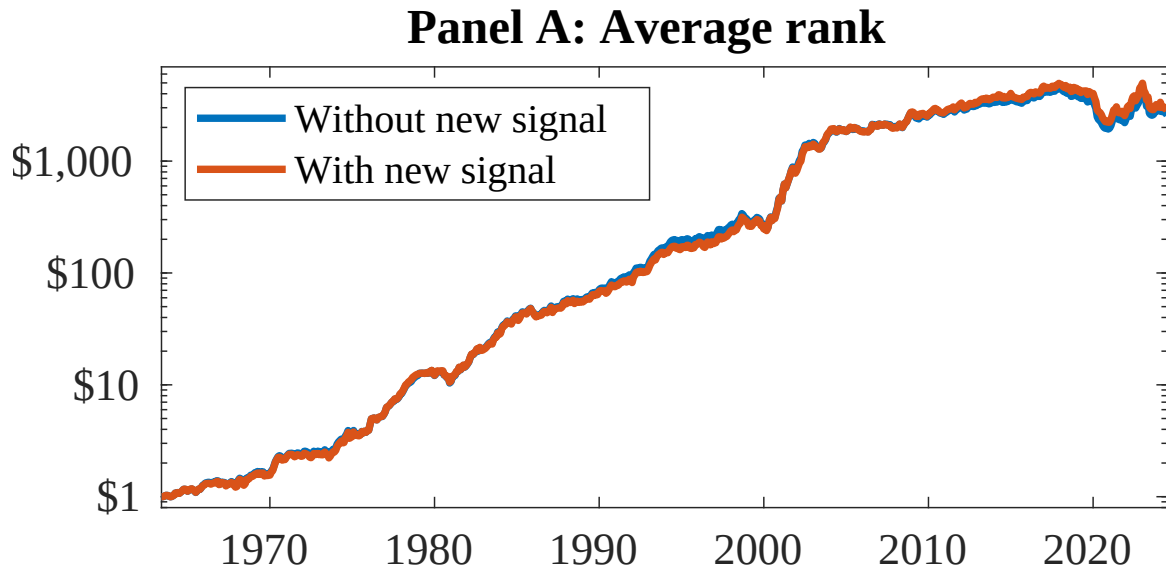


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CSD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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